

Package #175

Embed the enhanced physics-informed neural network model into the aircraft lithium battery management system to continuously predict cell surface temperature during flight and trigger protective actions (e.g., load reduction, active cooling) when predicted temperature approaches runaway thresholds.

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented problem 'thermal runaway' concerning aircraft lithium battery installation. User-submitted engineering problem (session input): "Why do aircraft lithium-battery installations suffer thermal events? (connector outage honesty)". [nasa_ntrs] 1 records retrieved live (2019-2019). Illustrative records: 'Statistical Characterization of 18650-Format Lithium-Ion Cell Thermal Runaway Energy Distributions'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. [doe_osti] 10 records retrieved live (2026-2026). Illustrative records: 'Mechanical abuse poses a critical safety risk to Li-ion batteries by inducing internal short circuits that initiate thermal runaway. Remarkably, voltage recovery frequently emerges during thermal runa'; 'The pursuit of higher energy density in lithium-ion batteries has made high-nickel (Ni) layered oxides leading cathode candidates for next-generation electric vehicles. However, their poor thermal sta'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

Mechanism: Incorporation of Joule heating, reversible entropic heat, and radiative losses into a physics-informed neural network for real-time temperature prediction.

Contents:

- 01 - Executive Technology Brief
- 02 - Engineering Technology-Transfer Dossier
- 03 - Buyer Decision Card
- 04 - Evidence Summary
- 05 - Transfer Manifest

Status: ENGINEERING_DEFINITION. Not physically validated.